

MAJOR CONTRIBUTORS TO THE STORY



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when compared to normal fluorescent lighting or even high-pressure sodium vapour lights for outdoor lighting," says Dhawan. He adds, "But the pay-back period, which is between two to three years, is so attractive that for any new installation, people are ready to shell out money on LED installation rather than the conventional lighting." Majority of the new installations, be it residential or commercial, are opting for LED lighting today.

Another aspect of this trend is the retro-fitting or replacement of LEDs. There are a lot of campaigns and activities that create awareness on the benefits of LED, even while replacing CFLs or other lighting. "Retro-fit is little slower as the initial cost is higher due to replacement of old fixture with new ones for LEDs," informs Dhawan. The latter is a slower process comparatively, but both are catching on.

Pricing trends

Despite the fact that LED provides huge energy-saving, LED market is not as large in India as it should be. This is mainly due to the fact that cost of LED lighting is much higher than that of traditional light bulbs. Raushan says, "Due to recent development with regard to manufacturing of LEDs in India, improvement in technology and competition, price of LEDs has reduced to a large extent. It has been estimated that by the next two to three years, price of LED light bulbs will be reduced by 50 per cent, but still its cost will be higher than that of traditional light bulbs."

However, with awareness about

energy-saving, application of LED lighting has increased, thus pushing down prices of LED light bulbs. "In India, in 2010, one LED lamp cost approximately ₹ 1200, which now in 2014, costs around ₹ 400," informs Raushan. It has been observed that, with the development and improvement in the manufacturing technology, cost of LED light bulbs has decreased almost 30 per cent in the last couple of years. This has also forced the local manufacturers to develop low-cost LED light bulbs.

Singh notes, "The price of an LED bulb of 5-7 watts, with power factor greater than 0.9 and total harmonic distortion lesser than 20 per cent, which was ₹ 400 in 2012, has reduced by 50 per cent." He adds, "Products that were once available for more than ₹ 400 in 2012 are now available for 50 per cent less." Luminaries in other categories or application areas have witnessed a similar reduction in price in India.

Dhawan believes, "Within a year or two, we should see the price of an LED lamp almost comparable to a normal CFL lamp or fixture."

Recommendation from the industry

Government is actively participating in promoting the LED segment, notes Dhawan. Street-lighting is one of the major applications of electric power in India. He says, "By helping municipalities of the civic authorities to have this perception and mindset change of transitioning from energy-guzzling, high-voltage sodium lights to LED, which at one-fourth the wattage can give you the kind of lighting experi-

ence in terms of light availability on the roads is actually a key area for the government to work upon."

Dhawan also says, "We expect the government to come up with regulations to help the Indian LED industry to set-up manufacturing of these products in India."

Lahiri believes, "There is still need for a stronger institutional mechanism. It is necessary to enable testing of LED products through the development of common testing protocols."

LED will need government support and R&D facilities to boom in future

Development of LED lighting market in India is primarily dependent on government and industry bodies, at least for the short-term duration. Once there is price rationalisation, demand for LED lights is expected to get a big boost, believes Raushan. He says, "Industry needs to develop its own R&D facilities, which help in the development of electronic components for LED lights. This is expected to reduce the cost of manufacturing." He adds, "Industry also needs to tackle sub-standard LED lights procured from Chinese market, which is hampering the LED market." This develops a negative marketing with regard to LED lighting systems. Industry should work toward reducing the import of materials required for the development of LED lights.

Lahiri also believes that the industry should look toward building strategic partnerships to obtain quality input at the right price. Developing an efficient distribution network, that can handle different segments such as enterprises, consumers and government bodies, would help the industry grow. He says, "Another important step is brand-creation in order to influence enterprises (B2B) and consumers (retail sales) for sales." Engaging in the development of innovative designs and improvements in product efficacy would give a push to the Indian LED market. ●

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